RRST Valve Controller — Hardware Modification Checklist

TMC2209 UART wiring + driver power-cycling protection, applied to the existing Teensy 4 board

Reference schematic: `RRST_eagle_schematics.sch` (updated with parts D1, D2, C7, C8, R1) · Prepared 2026-09-22

1. Parts needed

- ☐ 1× BigTreeTech TMC2209 V1.2 driver module (legacy-pinout / drop-in)
- ☐ 1× ~1 kΩ resistor (UART TX series resistor)
- ☐ 2× short jumper wires / 0Ω links (CFG1/CFG2 ground fix)
- ☐ 2× hookup wire (UART RX/TX taps)
- ☐ 1× 220 μF electrolytic capacitor, ≥35V, radial — **C7**
- ☐ 1× 100 nF ceramic capacitor — **C8**
- ☐ 1× SS34 Schottky diode (3A/40V axial, or equivalent) — **D1**
- ☐ 1× SMBJ26A TVS diode (26V standoff, axial/SMA) — **D2**
- ☐ 1× 10 kΩ resistor — **R1**

2. Part A — TMC2209 UART wiring

Enables software control of run current and microstepping over UART, in addition to normal STEP/DIR operation.

1. Insert the TMC2209 module into the existing `TMC2100`-silkscreened socket, in the standard orientation used by any drop-in A4988/TMC2100/TMC2208/TMC2209 stepstick (module's printed `TOP`/label side facing the same way the old driver did).
2. Sanity check only, no action: the module's `VCC_IO` pin lands on the socket's `VDD` pad, already wired to the Teensy's 3.3V rail.
3. Solder a wire from the socket's `MS3` pad (top row, 5th pin from the left) directly to **Teensy pin 0 (RX1)**. No resistor on this one.
4. Solder a second wire from that **same `MS3` pad** through a **~1 kΩ resistor** to **Teensy pin 1 (TX1)**. Both UART wires land on `MS3` — do not use the `!RST` pad (see note below).
5. Bridge the `CFG1` (`MS1`) pad to an adjacent GND pad with a short jumper wire.
6. Bridge the `CFG2` (`MS2`) pad to an adjacent GND pad with a short jumper wire.

> Steps 5–6 matter: `CFG1` / `CFG2` are currently unpopulated (floating), and the driver needs `MS1` = `MS2` =LOW to reliably latch UART address 0 at power-up.

> **Why both wires go to `MS3`, not split across `MS3` / `!RST`:** BTT's manual labels both pad positions `PDN_UART`, implying the module ties them together internally. Bench testing found that's not true for this module — measured resistance between the two pads was ~300 kΩ, not the ~1 kΩ expected if they shared a net. Landing both wires on the single confirmed-good `MS3` pad avoids depending on that assumption. If wiring a different module where this behaves differently, verify with a meter (power off) before trusting either pad in isolation.

3. Part B — Power-cycling protection

Addresses the driver damage seen on disconnect/reconnect: no bulk capacitance on the motor rail, no reverse-polarity/surge protection, and a floating `EN` pin during power transients.

1. Locate the `VMOT` pad at the driver socket and an adjacent `GND` pad.
2. Solder **C7** (220 μF/35V electrolytic) across `VMOT` / `GND`, as close to the socket as possible. **Mind polarity:** + to `VMOT`, – to `GND`.
3. Solder **C8** (100 nF ceramic) in parallel with C7, across the same two pads.
4. Identify the barrel jack's center-pin (`PWR`) trace to `VMOTOR` and cut/break it at a convenient point.
5. Splice in **D1** (SS34 Schottky) in series: jack side → anode, `VMOTOR` side → cathode.
6. Solder **D2** (SMBJ26A TVS) across `VMOTOR` / `GND`, near the jack or driver socket. Cathode to `VMOTOR`, anode to `GND`.
7. Solder **R1** (10 kΩ) between the `EN` pad (socket `EN` / Teensy pin 8 net) and a nearby `VCC` (3.3V) pad.

> **Diode polarity:** if D1 is reversed, the board simply won't power up (safe failure) — it won't damage anything, but double-check orientation before assuming a fault elsewhere.

> **Why SMBJ26A, not a higher-standoff part:** the TMC2209's real absolute maximum motor-supply rating is only 32–33V (checked directly against the datasheet's Absolute Maximum Ratings table) — much tighter than the TMC2100's 46V. No standard TVS diode can guarantee clamping under ~33V while keeping its standoff safely above the 24V rail (SMBJ26A itself clamps at ~42V under a full-rated pulse). SMBJ26A is the best practical choice, but treat it as a best-effort supplement — **C7 (the bulk capacitor) is the primary defense** against the reconnect/inductive-kickback transients this protection targets, since it works by slowing the voltage rise rather than clamping a peak.

4. Verification before first power-up

- [ ] Continuity check: `MS1` and `MS2` pads read 0Ω to GND
- [ ] `EN` pad is *not* shorted to GND (should read a pull toward 3.3V, high impedance)
- [ ] C7 / C8 polarity correct (visual)
- [ ] D1 orientation allows current jack → VMOTOR, not reversed (visual)
- [ ] D2 is SMBJ26A (not SMBJ33A), cathode toward VMOTOR, anode toward GND (visual)
- [ ] Power off: no short between VMOTOR and GND (multimeter resistance check)
- [ ] Power on: Teensy boots normally
- [ ] Once firmware is flashed: serial CLI confirms TMC2209 UART link detected

5. Software bring-up: current tuning & diagnostics (`p` / `x`)

Once firmware is flashed and the driver is enabled, two CLI commands help verify the build and address the motor overheating this project has seen (Motech MT-1705HS200AE, rated 2.0A — the driver's default hold current is 50% of run current, held continuously whenever the stepper is enabled but stationary, e.g. for as long as the tube clamp stays closed).

1. Send `x` — expect every field to read OK/ok, with no `WARNING: TMC2209 UART link is down` line and no `FAULT` / `SHUTDOWN` anywhere. If the link warning appears, re-check the UART wiring from Part A before trusting anything else on the board.
2. With the clamp closed and holding, send `p` and try progressively lower hold-current percentages (default is 50%). Check whether the clamp still holds against expected back-pressure from the tube at each value.
3. Find the lowest hold percentage that reliably holds, and note it — it resets to 50% on every reboot. If a bench-tested value proves reliably better, update the `DefaultHoldPercent` constant in the firmware so it isn't lost.
4. Send `x` again periodically during extended runs if overheating is suspected — a real electrical fault (short, overtemperature shutdown) shows up here, distinct from a mechanical/thermal issue that just needs a lower hold percentage.

> **Fault flags latch:** once `x` reports a fault, it stays reported until the stepper is disabled and re-enabled (`d` then `e`) — so `x` reflects "has anything gone wrong since last enable," not only the instant you check it.

6. Reference documents

Updated schematic: `RRST_eagle_schematics.sch`
Design specs: `docs/superpowers/specs/2026-09-22-tmc2209-uart-config-design.md` , `2026-09-22-driver-power-protection-design.md` , and `2026-09-23-hold-current-and-diagnostics-design.md`